

A Reproducible Computational Framework for Corpus-Wide Morphological Evaluation of the Voynich Manuscript

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Abstract

This study proposes a reproducible computational framework for evaluating a Latin-derived morphological hypothesis about the Voynich Manuscript as a compressed botanical-medical text. Rather than treating Voynich glyphs as simple character-substitution symbols, the proposed model represents EVA word tokens as structured combinations of reusable Latin-derived morphological components consisting of optional prefixes, lexical roots, and optional suffixes. The objective is not to claim a definitive decipherment, but to establish a falsifiable framework through which the proposed hypothesis can be systematically tested across the complete corpus.

A rule-based morphological parser was developed and evaluated using the complete EVA transcription of the Voynich Manuscript (31,007 tokens). Two complementary evaluation modes are distinguished. The Exploratory Recognition Mode identifies recurring morphological components and reproduces the earlier broad-coverage result of 30,590 recognized tokens (98.66% of the corpus). The Strict Full-Token Validation Mode, adopted as the primary evaluation criterion in this study, accepts a token only when its complete EVA sequence is fully explained by the proposed morphological grammar. Under this stricter criterion, the parser successfully accounts for 15,848 tokens (51.11% of the corpus), while maintaining consistent performance across all major manuscript sections, ranging from 45.7% to 57.8%.

Statistical validation was performed using 10,000 randomized lexical models evaluated under the identical strict parsing criterion. The randomized models produced a null mean of 2.112%, a standard deviation of 1.629 percentage points, and a maximum null result of 15.900%. The observed strict parsing rate corresponds to $z = 30.07\sigma$, with no randomized trial reaching the observed result (empirical $p < 0.0001$). Representative parsing examples further demonstrate that recurring morphological structures are reconstructed consistently across independent folios.

The proposed hypothesis is further evaluated by comparison with independent iconographic correspondence across the botanical, biological, and astronomical sections, including the reinterpretation of folios f99–f102 as plant anatomical diagrams, the biological section as representations of vascular flow, and the astronomical section as a seasonal botanical calendar. These visual correspondences do not constitute primary evidence but provide independent consistency checks on the explanatory scope of the proposed computational framework.

Taken together, these results establish a transparent, reproducible, and falsifiable computational framework for evaluating the Latin-derived morphological hypothesis as applied to the Voynich

Manuscript. The parser implementation, lexical resources, and evaluation datasets are openly released to facilitate independent reproduction and comparative evaluation of the proposed computational framework.

2. Introduction

2.1 The Voynich Manuscript as an Unresolved Problem

The Voynich Manuscript (Beinecke MS 408) remains one of the most extensively studied yet unresolved manuscripts in medieval scholarship. Since its rediscovery in the early twentieth century, the manuscript has attracted sustained attention from historians, linguists, cryptographers, computer scientists, and manuscript specialists. Despite more than a century of interdisciplinary investigation, no proposed interpretation has achieved broad scholarly acceptance, making the manuscript one of the most persistent open problems in the study of historical writing systems.

Radiocarbon dating places the parchment between approximately 1404 and 1438, confirming the manuscript as an authentic fifteenth-century artifact rather than a modern fabrication (University of Arizona, 2011). While its authorship, provenance, and intended purpose remain uncertain, the manuscript exhibits a coherent internal organization. Modern scholarship conventionally divides its contents into botanical, biological, astronomical, pharmaceutical, and recipe-like sections according to their dominant illustrations. These descriptive categories provide a practical framework for discussing the manuscript, although the relationship between the illustrations and the accompanying text remains unresolved.

Over the past two decades, computational studies have substantially advanced understanding of the manuscript's statistical structure. Analyses based on the Extensible Voynich Alphabet (EVA) have shown that the text exhibits several properties commonly associated with natural language, including structured word-frequency distributions, positional regularities, and non-random repetition patterns (Montemurro & Zanette, 2013). At the same time, recent linguistic analyses have noted that the manuscript resists straightforward classification as an ordinary natural language (Bowerman & Lindemann, 2021), reinforcing the need to evaluate alternative linguistic models.

The adoption of EVA has also improved the reproducibility of computational research by providing a common transcription standard (Landini & Zandbergen, n.d.). As a consequence, recent studies have increasingly emphasized corpus-level analyses rather than isolated examples, allowing competing hypotheses to be evaluated against the manuscript as a whole. These developments have significantly strengthened the empirical foundation of Voynich research, even though they have not resolved the relationship between the manuscript's textual structure and its visual organization. The

present study addresses one aspect of this broader question by examining whether both can be evaluated within a reproducible computational framework.

2.2 Previous Research and Outstanding Questions

Research on the Voynich Manuscript has generally followed four broad directions: historical and manuscript studies, cryptographic approaches, statistical and computational analyses, and linguistic decipherment. Each has contributed valuable insights while leaving important questions unresolved.

Historical scholarship has clarified the manuscript's provenance, dating, codicology, and relationship to medieval intellectual traditions (University of Arizona, 2011). Manuscript historians have also consistently interpreted its imagery within broader botanical and medical traditions, despite uncertainty regarding the identity of many illustrated plants (Zandbergen, 2025).

Cryptographic research has explored whether the manuscript represents an encrypted natural language or another form of encoded communication. Computational and statistical approaches have reported language-like regularities while also demonstrating that statistical structure alone cannot distinguish meaningful linguistic organization from alternative generative mechanisms (Rugg, 2004; Schinner, 2007; Montemurro & Zanette, 2013).

Linguistic decipherment proposals have explored a variety of underlying languages and encoding systems. Although several proposals have stimulated discussion, none has yet achieved broad acceptance as a reproducible corpus-wide explanation. Collectively, these studies suggest that an important remaining challenge is not only proposing plausible interpretations, but also establishing transparent methods through which competing hypotheses can be evaluated consistently.

2.3 Objectives of the Present Study

The present study investigates one possible explanation for the linguistic structure of the Voynich Manuscript based on reusable Latin-derived morphological components. Rather than attempting a direct alphabetic substitution or presenting isolated translations, the study evaluates whether recurring EVA tokens can be interpreted systematically as combinations of prefixes, lexical roots, and suffixes that recur throughout the manuscript.

The proposed computational framework combines corpus-wide rule-based parsing, conservative token-level validation, statistical comparison against randomized lexical models, and complementary iconographic evaluation. Under this framework, parser version v22 demonstrates substantial corpus-wide coverage under a conservative validation criterion and performs significantly better than randomized

lexical baselines. These results motivate further evaluation rather than constituting a definitive decipherment.

2.4 Research Hypothesis

The hypothesis evaluated in this study is that recurring Voynich word tokens encode reusable Latin-derived morphological components associated primarily with botanical and medical terminology. The model treats complete morphological tokens, rather than individual glyphs, as the principal unit of interpretation. Its theoretical basis is presented in the following section.

The hypothesis is intended to be empirically testable rather than interpretively self-validating. In particular, it would be weakened if randomized lexical models achieved comparable parsing performance under the same evaluation procedure or if the proposed model could not be applied consistently across independent sections of the manuscript.

The empirical criteria by which this hypothesis may be supported or challenged are presented in Section 3.7.

2.5 Origin of the Present Study

The study originated from repeated observations of folios f99–f102, conventionally assigned to the pharmaceutical section. Their layered internal organization and irregular outlines appeared difficult to reconcile with conventional vessel iconography, prompting consideration of an alternative anatomical interpretation and a broader search for recurring botanical terminology within the surrounding text.

Subsequent observations in the biological and astronomical sections encouraged the extension of this investigation to the manuscript as a whole. Throughout this study, iconographic observations are treated as complementary evidence rather than the primary basis for linguistic interpretation.

2.6 Contributions of the Present Study

This study makes five principal contributions: (1) it proposes a Latin Morphological Encoding Model; (2) it develops a reproducible computational evaluation framework; (3) it distinguishes exploratory recognition from strict token-level validation; (4) it evaluates the proposed hypothesis statistically against randomized lexical models; and (5) it integrates complementary iconographic correspondence into the overall evaluation strategy.

2.7 Organization of the Paper

Section 3 presents the theoretical framework. Section 4 describes the methodology. Section 5 reports the experimental results. Section 6 discusses their implications, historical context, and limitations. Section 7 concludes the study and outlines directions for future work.

3. Theoretical Framework

3.1 Research Perspective

The present study approaches the Voynich Manuscript as a problem of reproducible linguistic modelling rather than isolated translation. The objective is not to demonstrate that individual readings appear plausible, but to evaluate whether a constrained morphological model can account for recurring textual structures across the manuscript under explicit and reproducible rules.

The proposed computational framework examines the hypothesis that recurring EVA tokens encode reusable Latin-derived morphological components. This hypothesis is evaluated computationally at the corpus level and is complemented, rather than driven, by comparison with the manuscript's visual organization. Throughout this study, corpus-wide morphological consistency constitutes the primary line of evidence, whereas iconographic correspondence is treated as independent complementary evidence.

The hypothesis is intended as a computational proposition rather than a historical assertion.

3.2 The Latin Morphological Encoding Model

The proposed model assumes that Voynich tokens are composed of reusable morphological elements rather than direct alphabetic substitutions. The canonical organization of a token is represented as:

$$\text{Token} ::= (\text{Prefix}) \text{Root} (\text{Suffix})$$

This expression represents the canonical form of the model rather than a complete grammatical specification. Every valid token must contain at least one lexical root, while prefixes and suffixes may be absent, repeated, or combined according to the parser grammar described in Section 4.

Lexical roots are derived from Latin and Latinized botanical-medical terminology. Prefixes and suffixes modify or qualify these roots while remaining reusable across the corpus. Interpretation therefore operates at the morphological rather than the character level, and a successful interpretation requires that the complete token be explained without context-specific exceptions.

3.3 Interpretive Principles

Interpretation is performed at the token level rather than through isolated glyphs or partial substrings. Exploratory recognition is used only to identify candidate morphological components during lexical development, whereas formal evaluation requires complete token-level explanation under a fixed rule set.

A proposed interpretation is considered theoretically meaningful only when it is structurally compatible with the morphological grammar, supported by the lexical knowledge base, and reproducible across independent occurrences. The computational implementation of these principles is presented in Section 4.

3.4 Fixed Morphological Values

Morphological components retain stable semantic values within a given parser version. Prefixes, roots, and suffixes are not redefined from one context to another to accommodate individual examples. Instead, version changes are applied globally between documented parser versions and are subsequently re-evaluated across the complete corpus.

Stable lexical values are essential for reproducibility because they allow independent researchers to determine whether identical rules produce comparable interpretations across unrelated manuscript sections. The proposed computational framework therefore treats semantic stability as a methodological constraint. Its empirical evaluation is reported through corpus-wide lexical stability analysis in the Results section.

3.5 Lexical Families and Orthographic Variants

Fixed lexical values do not imply that every graphical realization must correspond to a unique lexical assignment. Instead, the model distinguishes between **lexical families** and their possible **orthographic variants**. A lexical family represents a stable semantic unit within the parser, whereas orthographic variants represent alternative surface realizations that are treated as belonging to the same family when supported by corpus-wide evidence.

This definition is operational rather than historical and is adopted solely for computational evaluation.

The recognition of orthographic variants is rule-based rather than ad hoc. Variants are documented within the lexical knowledge base, remain fixed throughout a parser version, and are

evaluated globally across the corpus. No new variants are introduced to explain individual tokens during formal evaluation.

This distinction allows the model to accommodate systematic graphical variation without abandoning the principle of fixed lexical values. The empirical distribution of lexical families and their stability across manuscript sections is evaluated in the Results section.

3.6 Functional and Lexical Morphemes

The proposed model distinguishes between **lexical morphemes**, which contribute the primary semantic content of a token, and **functional morphemes**, which perform grammatical or relational functions. Functional morphemes are generally shorter and frequently recurring, whereas lexical morphemes carry the principal botanical, anatomical, or procedural information.

Because short functional elements occur widely throughout the corpus, they are never evaluated in isolation. Their contribution is assessed only within complete-token parsing under the strict grammatical constraints of the parser. Consequently, parser performance depends on successful explanation of the entire token rather than on the presence of individual short components.

This distinction limits overinterpretation of highly frequent short forms while preserving their legitimate grammatical role within the proposed model.

3.7 Iconographic Correspondence

Within the proposed computational framework, the manuscript's illustrations serve as independent complementary evidence rather than as the primary basis of linguistic interpretation. Linguistic analyses are generated first and subsequently compared with their visual context. Iconographic observations are not used to define lexical values.

This comparison is examined across the botanical, biological, astronomical, and pharmaceutical sections, including the reinterpretation of folios f99–f102. Consistent correspondence strengthens the overall explanatory coherence of the hypothesis, whereas systematic disagreement weakens it.

3.8 Empirical Evaluation Philosophy

The proposed computational framework is intended to support empirical testing through explicit evaluation criteria.

Two criteria determine whether the proposed hypothesis is supported. First, it should perform substantially better than randomized lexical models under identical parsing procedures. Second, the same

model should generalize across independent manuscript sections rather than being restricted to selected examples.

These criteria motivate the distinction between exploratory recognition and strict full-token validation. Exploratory recognition supports hypothesis development, whereas strict validation provides the principal criterion for formal evaluation.

Model refinement follows a separate governance principle. Version changes are expected to be explicitly documented, applied globally rather than locally, and re-evaluated against the complete corpus. The detailed implementation of parser construction, version management, and evaluation procedures is presented in Section 4.

Collectively, these principles establish the conceptual separation between historical interpretation and computational evaluation that underlies the remainder of this study.

4. Methodology

4.1 Research Design

The objective of this study is not to produce isolated lexical interpretations but to evaluate whether the proposed Latin-derived morphological model can account for recurring textual structures throughout the Voynich Manuscript under explicit, reproducible conditions. The methodological workflow consists of six stages:

1. Corpus preparation
2. Construction of a version-controlled lexical knowledge base
3. Rule-based parser implementation
4. Corpus-wide validation
5. Statistical evaluation
6. Version governance and reproducibility

The sixth stage documents version control and reproducibility, ensuring that every reported result can be independently reproduced from a fixed parser version.

Rather than maximizing apparent decipherment, the methodology adopts conservative validation. Tokens that cannot be explained under explicit parser constraints remain unresolved rather than being interpreted post hoc.

4.2 Corpus Preparation

The complete EVA (Extensible Voynich Alphabet) transcription was used as the textual corpus. EVA was selected because it is the most widely adopted transcription standard in computational Voynich research.

Preprocessing consisted of:

- transcription normalization
- token extraction
- positional indexing
- section classification
- duplicate preservation

No manual correction or selective token removal was performed.

The manuscript was analyzed both as a complete corpus and by four major illustrated groups:

- Botanical
- Biological
- Astronomical
- Pharmaceutical (including folios f99–f102)

Importantly, these section labels were used only during post-parsing analysis. They were explicitly excluded from parser execution in order to prevent circularity between linguistic interpretation and iconographic classification.

The corpus itself remained unchanged throughout development. Only the lexical knowledge base and parser versions evolved.

4.3 Construction of the Lexical Knowledge Base

The parser operates on a version-controlled lexical knowledge base composed of reusable Latin-derived morphological components rather than whole-word translations.

Each lexical entry records:

- morphological category
- proposed Latin-derived value
- semantic function
- parser status

4.3.1 Inclusion Criteria

Candidate components were admitted only when they:

7. possessed documented Latin or Latinized botanical-medical origins;
8. recurred sufficiently to support corpus-level evaluation while contributing to multiple independent token interpretations;
9. remained compatible with the parser grammar;
10. improved corpus-wide consistency without context-specific semantic reassignment.

4.3.2 Fixed Lexical Values

Within a given parser version, lexical values remain fixed throughout corpus-wide evaluation. Context-dependent reassignment is prohibited.

When version changes became necessary, they were introduced only between documented parser versions and applied globally before the entire corpus was reprocessed.

4.3.3 Lexical Family Assignment

Orthographic variants were assigned to the same lexical family only when their relationship had been documented in the lexical knowledge base prior to corpus-wide evaluation. Such groupings were based on stable graphical correspondences observed across multiple independent tokens and were treated as fixed parser definitions rather than post hoc interpretive adjustments.

Lexical family membership was established only when candidate variants shared a documented core graphical segment and differed exclusively by predefined orthographic or affixal variation recorded in the lexical knowledge base. Semantic similarity alone was never considered sufficient for assigning two forms to the same lexical family.

Consequently, lexical family membership remained fixed throughout parser version **v22** and was never modified to improve the interpretation of an individual token or manuscript section. This procedure distinguishes documented orthographic variation from semantic reassignment and ensures that improvements in parser performance cannot arise from retrospective regrouping of graphical variants after corpus-wide evaluation.

This definition applies uniformly throughout parser version **v22** and underlies all lexical-family analyses reported in the Results and Discussion chapters.

4.3.4 Version Control

All results reported in this study were generated using parser version **v22**, representing a frozen lexical inventory and deterministic evaluation state.

4.4 Parser Architecture

The parser evaluates complete EVA tokens using deterministic rule-based parsing.

4.4.1 Parsing Grammar

Token ::= (Prefix) *Root* (Suffix)

Every valid interpretation must contain at least one lexical root. Prefixes and suffixes may be absent, repeated, or combined according to the parser grammar.

4.4.2 Parsing Workflow

For each token the parser:

11. normalizes the EVA token;
12. identifies candidate lexical roots;
13. identifies compatible prefixes and suffixes;
14. generates valid grammatical segmentations;
15. evaluates candidate interpretations;
16. returns the highest-ranked valid interpretation or classifies the token as unresolved.

4.4.3 Candidate Ranking

When multiple valid interpretations exist, deterministic ranking rules are applied.

Priority is given to:

17. complete explanation of the token;
18. minimal unmatched characters;
19. minimal unnecessary segmentation;
20. deterministic tie-breaking.

Consequently, identical parser versions always produce identical results for identical inputs.

4.4.4 Exploratory Recognition and Strict Full-Token Validation

Exploratory Recognition is used during lexical development to identify candidate morphological components.

Strict Full-Token Validation constitutes the formal evaluation reported in this paper. A token is counted as successfully parsed only when its complete structure is explained using fixed lexical values.

4.4.5 Lexical Stability Enforcement

Parser implementation prevents context-specific semantic reassignment.

Lexical refinements require creation of a new parser version followed by complete corpus-wide re-analysis.

4.5 Validation Strategy

Validation was performed at token, corpus, and lexical levels.

4.5.1 Token-Level Validation

Only complete-token interpretations produced under Strict Full-Token Validation were included in reported performance.

4.5.2 Corpus-Level Validation

Corpus-wide evaluation included overall coverage, section-specific coverage, unresolved tokens, and distribution across manuscript sections.

4.5.3 Lexical Stability Analysis

Lexical stability was evaluated independently of parsing coverage.

Within each parser version:

- lexical values remained fixed;
- no context-specific reassignment occurred during parsing;
- version changes occurred only between documented versions.

Quantitative results are presented in Section 5.

4.6 Statistical Evaluation

4.6.1 Null Model Construction

A permutation-based null model was constructed by generating randomized root inventories while preserving the parser architecture, corpus tokenization, prefix definitions, suffix definitions, and evaluation procedure. For each permutation trial, the number of randomized roots was set equal to the number of roots in the v22 lexical knowledge base. Random roots were sampled from the observed cleaned EVA vocabulary of the manuscript rather than generated as synthetic strings.

This design tests whether a randomly selected set of observed EVA token forms, when evaluated under the same strict parser grammar, can achieve parsing coverage comparable to the proposed Latin-derived root inventory. In strict mode, no partial substring fallback is permitted: a token is counted as parsed only when the full token is consumed as an exact root or as a valid combination of parser-defined prefix, root, and suffix.

4.6.2 Permutation Procedure

For each randomized root inventory, the identical parser was executed under Strict Full-Token Validation, and corpus-wide parsing coverage was recorded.

The permutation procedure was repeated **10,000** times to generate an empirical null distribution.

The observed parser performance was compared with the null distribution using:

- null mean
- null standard deviation
- maximum null performance
- empirical exceedance frequency
- standardized z-score

4.7 Reproducibility and Version Governance

Reproducibility was a primary design objective.

Parser version changes followed three governance principles:

21. every version change was explicitly documented;
22. version changes were applied globally rather than locally;
23. every version change required complete corpus-wide re-evaluation.

All reported results correspond to parser version **v22**.

The parser implementation, lexical knowledge base, evaluation scripts, and supporting resources have been released through GitHub and archived via Zenodo to facilitate independent reproduction and future refinement.

Every result reported in this study can therefore be reproduced from a fixed parser version using the publicly released parser, lexical knowledge base, and evaluation scripts without modification of parser rules or lexical values.

5. Results

5.1 Corpus-wide Parsing Performance

The parser was evaluated against the complete EVA corpus of the Voynich Manuscript, consisting of **31,007** tokens. Two evaluation modes were reported: Exploratory Recognition and Strict Full-Token Validation. Exploratory Recognition measures broad morphological component recognition, whereas Strict Full-Token Validation accepts only tokens whose complete EVA sequence is explained by the fixed parser grammar.

Table 1. Corpus-wide parsing performance

Metric	Count	Percentage
Total tokens	31,007	100.00%
Exploratory Recognition	30,590	98.66%
Strict Full-Token Validation	15,848	51.11%
Unresolved tokens	15,159	48.89%

The difference between the exploratory and strict results is expected. Exploratory Recognition was used to identify recurring candidate components during model development, while Strict Full-Token Validation was used as the primary evaluation criterion. The strict result therefore represents a conservative estimate of corpus-wide morphological coverage under the proposed grammar.

5.2 Section-Level Consistency

Strict parsing performance was evaluated across the principal manuscript sections. Section labels were not used during parser execution; they were applied only after parsing to evaluate whether performance generalized across visually distinct manuscript groups.

Table 2. Section-level strict parsing consistency

Section	Total Tokens	Strict Parsed	Strict Parsing
Botanical	11,080	5,363	48.40%
Biological	6,872	3,964	57.68%
Astronomical	283	139	49.12%
Pharmaceutical (Roots)	984	450	45.73%
Pharmaceutical (Recipes)	906	524	57.84%
Recipes	10,882	5,408	49.70%

Strict parsing performance ranged from **45.73%** to **57.84%** across the reported sections. This relatively narrow range is important because the parser was not optimized separately for individual visual categories. The results therefore support the claim that the model is not limited to a single manuscript section or to the folios from which the hypothesis originated.

The highest strict parsing rate was observed in the Pharmaceutical (Recipes) group, while the lowest was observed in Pharmaceutical (Roots). This variation is discussed further in Section 6, especially in relation to the iconographic reinterpretation of folios f99–f102.

5.3 Statistical Validation

The strict parsing result was evaluated against a permutation-based null model. In each trial, a randomized root inventory was sampled from the observed cleaned EVA vocabulary while preserving the parser architecture, prefix definitions, suffix definitions, tokenization, and strict evaluation procedure.

Table 3. Permutation test statistics

Metric	Value
Permutation trials	10,000
Null mean	2.112%
Null standard deviation	1.629 percentage points
Maximum null performance	15.900%
Observed strict parsing	51.11%
z-score	30.07 σ
Trials $\geq$ observed	0 / 10,000
Empirical p-value	< 0.0001

The observed strict parsing rate exceeded the null mean by approximately **24.2 times**. No randomized trial reached the observed strict parsing rate, and the maximum null result (**15.900%**) remained far below the observed value (**51.11%**). These results indicate that the proposed Latin-derived

root inventory performs far beyond what was observed from randomly sampled root inventories under the same strict parser conditions.

5.4 Representative Morphological Parsing

Representative strict parses were extracted directly from the parser output without manual modification. These examples illustrate how tokens from multiple manuscript sections are analyzed as combinations of prefixes, roots, and suffixes.

Table 4. Representative strict parsing examples

Folio	Section	EVA Token	Prefix	Root	Suffix	Latin Value	Interpretation
f1r	Botanical	yshey	y	sh	ey	<i>suber</i>	demonstrative + cork/suber + from/at
f1r	Botanical	ycheey	y	ch	eeey	<i>cortex</i>	demonstrative + bark/cortex + emphatic of
f75r	Biological	qoldy	qo	l	dy	<i>il-</i>	quod/de + article + de
f75r	Biological	lshey	l	sh	ey	<i>suber</i>	article + cork/suber + from/at
f71r	Astronomical	rary	r	ar	y	<i>radius</i>	re- + medullary ray/radius + adjective
f73r	Astronomical	yteey	y	t	eeey	<i>tunica</i>	demonstrative + membrane/layer + emphatic of
f87r	Pharmaceutica l (Roots)	ykeey	y	k	eeey	<i>coquitur</i>	demonstrative + decoction/processing + emphatic of
f87r	Pharmaceutica l (Roots)	pcheos	p	ch	eos	<i>cortex</i>	pro + bark/cortex + seasonal plural
f99r	Pharmaceutica l (Recipes)	pcheody	p	cheo	dy	<i>cortex</i>	pro + bark/cortex + de
f99r	Pharmaceutica l (Recipes)	qokeeor	qo	k	eeor	<i>coquitur</i>	quod/de + decoction/processing + aureus/emphatic
f103r	Recipes	qoteey	qo	t	eeey	<i>tunica</i>	quod/de + membrane/layer + emphatic of
f103r	Recipes	opchedy	op	ch	edy	<i>cortex</i>	contra + bark/cortex + per

The examples shown in Table 4 represent documented lexical families rather than isolated root assignments. Orthographic variants belonging to the same lexical family retain a fixed semantic value

throughout parser version v22, as defined in Section 3.5. The parser therefore distinguishes orthographic variation from semantic reassignment.

The examples show that the same lexical values recur across independent sections. For example, *ch* / *cheo* is associated with *cortex*, *sh* with *suber*, *t* with *tunica*, *k* with *coquitur*, and *ar* with *radius*. These examples are not presented as isolated translations; rather, they illustrate the parser's rule-based decomposition under strict validation.

The full parser output is provided in Supplementary Table S2.

5.5 Iconographic Correspondence

Iconographic comparison was performed after linguistic parsing and did not influence parser execution. The purpose of this comparison was to evaluate whether the strict parser results were compatible with the manuscript's visual organization.

The representative examples in Table 4 show repeated botanical-anatomical values across visually distinct sections. In the botanical section, tokens containing *cortex* and *suber* occur in folios containing plant imagery. In the biological section, related anatomical terms occur in contexts visually dominated by channels and flow-like structures. In the astronomical section, terms such as *radius* and *tunica* appear in circular or layered visual contexts. In folios traditionally classified as pharmaceutical, including f99r, terms such as *cortex* and *coquitur* occur in proximity to forms that the present study evaluates as possible plant anatomical or processing diagrams.

These correspondences do not constitute primary evidence for decipherment. They function as independent complementary evidence: linguistic parses are first produced by the strict parser, and only afterward compared with visual context. The observed correspondences therefore support the explanatory coherence of the model without making iconographic interpretation circular.

5.6 Lexical Stability Analysis

The parser was evaluated for lexical stability by examining whether root values remained fixed across corpus-wide strict parsing. Table 5 summarizes representative root entries and their occurrence across manuscript sections.

Table 5. Representative lexical families and stability

Lexical Family	Variant	Latin Value	Interpretation	Occurrences	Section Count	Sections	Context-Specific Reassignment
cortex	ch	<i>cortex</i>	bark/cortex	2,056	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
coquitur	k	<i>coquitur</i>	decoction/processing	1,884	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
suber	sh	<i>suber</i>	cork/suber	1,023	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
duramen	daiin	<i>duramen</i>	heartwood	747	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
tunica	t	<i>tunica</i>	membrane/layer	669	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
lignum	ol	<i>lignum</i>	wood/lignum	649	5	Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
exterior	ok	<i>exterior</i>	outer layer	608	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical	No

Lexical Family	Variant	Latin Value	Interpretation	Occurrences	Section Count	Sections	Context-Specific Reassignment
exterior	ot	<i>exterior</i>	surface	596	6	(Roots); Recipes Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
annulus	aiin	<i>annulus</i>	growth ring	508	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
aureus	or	<i>aureus</i>	golden/yellow	394	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
cortex	chol	<i>cortex</i>	bark/cortex	383	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
radius	ar	<i>radius</i>	medullary ray	377	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
il-	l	<i>il-</i>	article	334	5	Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No
aut	o	<i>aut</i>	or/aut	320	6	Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical	No

Lexical Family	Variant	Latin Value	Interpretation	Occurrences	Section Count	Sections	Context-Specific Reassignment
est	s	est	is/est	319	6	(Roots); Recipes Astronomical; Biological; Botanical; Pharmaceutical (Recipes); Pharmaceutical (Roots); Recipes	No

The stability analysis directly addresses a potential objection to morphological decipherment: that component meanings might be redefined opportunistically according to context. Lexical stability is evaluated at the level of lexical families rather than individual graphical realizations. Multiple orthographic variants may belong to the same documented lexical family while preserving identical semantic values. Such relationships are established before corpus-wide evaluation and remain fixed throughout parser version v22. No context-specific semantic reassignment is introduced during strict parsing. This supports the methodological claim that the model is reproducible at the level of lexical values as well as token-level coverage.

The complete lexical stability output is provided in Supplementary Table S3.

5.7 Robustness Analysis: Short-Root Ablation

To evaluate whether the reported strict parsing performance was primarily driven by highly frequent short lexical roots, two controlled ablation experiments were performed after completion of the principal corpus analysis. These experiments temporarily removed subsets of the lexical root inventory while preserving tokenization, parser grammar, prefix definitions, suffix definitions, manuscript section assignments, and the Strict Full-Token Validation procedure. The official parser implementation therefore remained unchanged except for the temporary removal of selected lexical roots.

Table 6. Summary of short-root ablation experiments

Condition	Strict Parsed	Strict Coverage
Baseline parser (v22)	15,848	51.11%
One-character roots removed	12,038	38.82%
One- and two-character roots removed	6,317	20.37%

Removing all one-character lexical roots reduced strict coverage by **12.29 percentage points (24.04% relative)**. Extending the experiment to remove every one- and two-character lexical root reduced strict coverage by **30.74 percentage points (60.14% relative)** compared with the baseline parser.

Importantly, neither experiment eliminated successful strict parsing. Even after removing every one- and two-character lexical root, the parser continued to explain **6,317 corpus tokens (20.37%)** under identical Strict Full-Token Validation rules.

Notably, the second ablation also removed **ch**, the most frequently occurring lexical root in the corpus (**2,056 occurrences**) and one of the few lexical families represented across all six manuscript sections. Despite eliminating this dominant lexical family together with every remaining one- and two-character lexical root, substantial corpus-wide parsing performance remained. This observation argues against the interpretation that the reported strict coverage is produced solely by a small number of highly frequent short lexical elements.

The removed lexical roots are listed below to ensure complete reproducibility.

One-character roots: d, k, l, o, p, r, s, t

Additional two-character roots: al, am, ar, ch, dy, ee, od, ok, ol, om, or, ot, sh

5.8 Summary

The empirical evidence presented in this chapter supports the proposed framework through four complementary forms of validation:

- (1) Conservative corpus-wide performance under Strict Full-Token Validation (51.11%).
- (2) Statistical separation from randomized lexical inventories through permutation testing ($z = 30.07\sigma$; empirical $p < 0.0001$).
- (3) Consistent parser behavior across independent manuscript sections.
- (4) Robustness under progressively stricter lexical ablation, demonstrating that corpus-wide explanatory power persists even after removal of every one- and two-character lexical root.

Taken together, these analyses indicate that the reported parser performance cannot be explained solely by randomized lexical assignment, by manuscript section effects, or by the presence of highly frequent short lexical roots. Instead, they support the existence of a reproducible corpus-wide morphological structure under the constraints of the proposed parser framework.

6. Discussion

6.1 Interpretation of the Statistical Evidence

The principal result of this study is not a claim that the Voynich Manuscript has been deciphered. The central result is more limited, but also more reproducible: under a fixed lexical inventory, fixed tokenization, fixed section definitions, and a conservative Strict Full-Token Validation criterion, the v22 parser accounts for 15,848 of 31,007 evaluated tokens, or 51.11% of the corpus. This result is intentionally distinguished from the broader Exploratory Recognition Mode, which recognizes 30,590 of 31,007 tokens, or 98.66%, by allowing legacy substring recognition. The distinction is methodologically important. Exploratory recognition may be useful for hypothesis generation and for locating recurrent internal material, but the stricter measure is the primary validation criterion used for evaluating whether the proposed root inventory supports full-token parsing.

The strict result should therefore be interpreted as evidence for partial but reproducible morphological organization, not as evidence of complete translation. A model that parses approximately half of the corpus under conservative rules is not equivalent to a decipherment. At the same time, it is not trivial. The strict parser requires that a token be consumed as an optional prefix, a recognized root, and an optional suffix, with no substring fallback. Under this condition, performance remains far above the null expectation estimated by randomized lexical inventories. The permutation experiment reported in Chapter 5 evaluates the observed strict parser against random root sets of identical size using the same full-token criterion. The observed rate of 51.11% is separated from the null distribution by approximately 30.07 standard deviations, with no random trial reaching the observed rate in 10,000 trials. This result does not prove the historical interpretation of the system, but it does show that the selected lexical inventory is not behaving like an arbitrary set of strings under the same evaluation procedure.

The statistical evidence is strongest when read cumulatively rather than as a single number. Corpus coverage, section-level consistency, permutation testing, lexical stability, failure-case reporting, and robustness ablation experiments each constrain a different source of possible overinterpretation. The coverage figure asks how much of the corpus is parsed under a strict rule. The permutation test asks whether the same level of strict parsing could arise from random lexical inventories of the same size. The lexical-stability tables ask whether a root is reassigned opportunistically from one context to another. The failure-case tables make unresolved material visible instead of hiding it. The ablation experiments ask how dependent the result is on short lexical entries. Taken together, these tests support the conclusion that v22 captures a reproducible, non-random morphological signal while leaving open the larger historical question of what that signal represents.

This distinction between statistical structure and historical interpretation is essential for peer review. A reproducible parser may identify structure without yet explaining the origin, language, scribal convention, or intellectual context of that structure. Conversely, a historically attractive interpretation is not sufficient unless it survives computational constraints. The present study therefore treats the parser as a model-driven instrument: it formalizes a hypothesis, applies it consistently, reports both successes and failures, and exposes the model to controlled perturbations. The result is not a final reading of the manuscript, but a testable framework in which stronger or weaker versions of the hypothesis can be compared.

6.2 Lexical Families, Orthographic Variation, and Semantic Stability

A central methodological feature of v22 is the distinction between lexical families and orthographic variants. The parser does not require every EVA form to be treated as an independent semantic unit. Instead, it allows several predefined variants to belong to a stable lexical family when those variants share a fixed semantic value. For example, the variants *ch* and *chol* are treated as members of the *cortex* family, while *ok* and *ot* are treated as members of the *exterior* family. This does not mean that orthographic variation is ignored. Rather, orthographic variation is represented explicitly at the level of the variant field, while semantic stability is represented at the level of the lexical family.

This distinction matters because manuscripts commonly exhibit graphical, scribal, or abbreviated variation without requiring semantic reassignment at every occurrence. A short form and a fuller form may be related by abbreviation, expansion, positional convention, scribal habit, or graphical economy. In a computational model, however, such variation can easily become a source of overfitting if each form is allowed to acquire a new meaning whenever a local interpretation requires it. The v22 lexical-stability tables therefore separate the variant from the lexical family. The variant column preserves the actual parser root form, while the *lexical_family* column groups forms that share the same stable semantic value. This structure allows the paper to discuss families such as *cortex* or *exterior* without erasing the underlying EVA forms from which those families are inferred.

The distinction also clarifies the difference between orthographic variation and semantic reassignment. Orthographic variation occurs when multiple written forms are assigned to the same stable value. Semantic reassignment would occur if a given form were allowed to mean one thing in one section and another thing in another section. The v22 parser does not use context-specific reassignment within the fixed lexical inventory. Each root variant has a fixed Latin value, and the lexical-stability export marks context-specific reassignment as "No" for every row. This does not prove that every assignment is

historically correct, but it does prevent one of the most common failure modes in proposed decipherments: changing the meaning of a sign or word whenever the surrounding passage demands a different reading.

Lexical-family modeling therefore performs two functions at once. First, it permits a realistic amount of orthographic and abbreviatory variation. Second, it provides a constraint against arbitrary semantic drift. The same family assignment can be challenged, modified, or removed in future experiments, but such changes must be made in the lexicon and then evaluated across the entire corpus. This turns lexical disagreement into a reproducible computational comparison. A critic who rejects the cortex family, for example, can alter that family, rerun the parser, and measure the resulting change in strict coverage, section consistency, failure distribution, and permutation behavior. The proposed computational framework does not ask readers to accept lexical families by authority; it asks that disagreements be tested against the full corpus under identical rules.

The lexical-family approach also explains why the paper reports both representative examples and full supplementary outputs. Representative examples are useful for showing how a parsed token is decomposed into prefix, root, and suffix. They are not, by themselves, sufficient evidence. For this reason, the representative table is paired with the full token-level parser output and the full lexical-stability table. The examples illustrate the mechanism; the supplementary files allow the mechanism to be audited. This is especially important in a manuscript as contested as the Voynich Manuscript, where isolated readings have often been more persuasive than reproducible systems. The v22 methodology deliberately reverses that emphasis: individual examples are secondary to the full parser output.

6.3 Pharmaceutical Subsections and the Use of Parser-Assigned Categories

The distinction between Pharmaceutical (Roots) and Pharmaceutical (Recipes) requires explicit explanation because two classification systems are being used for different purposes. The broader discussion of the manuscript follows the traditional manuscript classification, in which the pharmaceutical material is often treated as a visually and codicologically recognizable region. This traditional category is useful when discussing iconography, layout, vessel-like or root-like drawings, plant parts, and the historical question of how the section should be understood within the manuscript as a whole.

The parser, however, assigns more specific subsections for computational reporting. In the exported tables, Pharmaceutical (Roots) corresponds to the folio range f87-f96v, while Pharmaceutical (Recipes) corresponds to f99-f102. This distinction is not intended to overturn the traditional classification. It is a methodological subdivision used to measure whether different parts of the traditionally pharmaceutical material behave differently under the same parser. The two subsections differ in layout, label density, and textual function. The f87-f96v material is treated as a root or plant-part

catalogue in the parser reports, while f99-f102 is treated as a recipe-like or part-labelled pharmaceutical subsection. Reporting them separately makes it possible to ask whether strict parsing succeeds uniformly across the pharmaceutical material or whether performance varies by internal function.

This distinction also explains why representative examples follow parser-assigned subsections while the broader discussion follows the traditional manuscript classification. A representative parsing table is a computational artifact. Its purpose is to show actual parser output from each evaluated section label. Therefore it uses the parser's section labels: Botanical, Biological, Astronomical, Pharmaceutical (Roots), Pharmaceutical (Recipes), and Recipes. These labels are used consistently in the exported CSV files so that every example can be traced back to the same sectioning rules used for token counts and section statistics.

The broader interpretive discussion has a different purpose. It asks how the manuscript's visual and textual organization might be understood historically. For that question, the traditional manuscript classification remains relevant, because the reader must be able to compare the proposed interpretation with established descriptions of the codex. Thus, when this chapter discusses the pharmaceutical material in a historical or iconographic sense, it may refer to the pharmaceutical section as a traditional manuscript category. When it discusses parser performance or representative examples, it follows the parser-assigned subsections. The two usages are not contradictory; they operate at different levels of analysis.

The distinction is also useful because it prevents circular argument. If the parser were allowed to redefine manuscript sections solely to improve the interpretation, section-level results would be difficult to evaluate. In v22, section classification is fixed and reported. The parser-assigned subsections are not hidden, and their token counts and strict parsing rates are exported. Pharmaceutical (Roots) contains 984 evaluated tokens with 450 strict parses, or 45.73%. Pharmaceutical (Recipes) contains 906 evaluated tokens with 524 strict parses, or 57.84%. These figures are not used to force a conclusion about the historical identity of the section. They provide a reproducible basis for discussing whether internally different portions of the pharmaceutical material show different degrees of compatibility with the proposed morphology.

6.4 Functional Morphemes, Short Roots, and Robustness

A major concern for any parser of the Voynich Manuscript is that apparent success may be driven by very short forms. Short roots are frequent, easy to match, and potentially vulnerable to overinterpretation. The proposed computational framework therefore includes robustness experiments designed to measure how much of the strict parsing result depends on short lexical entries. These experiments do not change the parser grammar, tokenization, prefixes, suffixes, section labels, or official

reported results. They temporarily remove selected ROOT entries from the lexical inventory and then rerun the same Strict Full-Token Validation logic.

The baseline strict result is 15,848 fully parsed tokens, or 51.11% of the 31,007-token evaluated corpus. In the first ablation experiment, all one-character ROOT entries were removed from the lexical inventory. Prefixes and suffixes remained unchanged, and the strict parser continued to require complete token consumption. Under this condition, strict coverage fell to 12,038 tokens, or 38.82%. This reduction shows that one-character roots contribute substantially to the current parser's coverage. The result is not surprising, because several one-character forms have high frequency and can participate in strict parses when combined with prefixes or suffixes. The experiment is valuable precisely because it quantifies that contribution rather than leaving it implicit.

The second robustness experiment removed all lexical ROOT entries whose EVA key length is one or two characters. This removed the one-character roots together with short two-character roots such as *ch*, *sh*, *ol*, *ok*, *ot*, *ar*, *al*, and related forms. Under the same strict parsing logic, coverage fell to 6,317 tokens, or 20.37%. This result confirms that short lexical roots are a major component of v22 performance. If all one- and two-character roots are removed, the parser loses more than half of its baseline strict coverage. The absolute drop from baseline is 30.74 percentage points, and the relative drop is 60.14%.

These findings are a strength of the methodology because they make model dependence visible. A weaker presentation would report only the baseline result and leave readers to wonder whether the parser is mainly a collection of short high-frequency matches. The ablation tests answer that concern directly. They show that short roots matter; they also show how much they matter under controlled conditions. The one-character removal reduces coverage but leaves 38.82% strict parsing. The one- and two-character removal is more severe, leaving 20.37% strict parsing. These values provide a quantitative basis for discussing the role of short lexical units rather than treating the issue as a rhetorical objection.

At the same time, the ablation results are also a limitation. They show that v22 is not yet a balanced explanatory model in which all lexical classes contribute equally. Short roots carry a large share of the present coverage. Some of this may be linguistically meaningful if the underlying system is abbreviatory, because compressed systems often rely on short high-frequency morphemes. Some of it may also reflect the danger inherent in modeling a corpus with a small alphabet and repeated glyph sequences. The current experiments cannot distinguish these possibilities by themselves. They show dependence; they do not yet determine whether that dependence is historically expected, scribally motivated, or partly artifactual.

The final implication is therefore methodological rather than conclusive. The current ablation experiments quantify the cumulative contribution of short lexical roots, but they do not yet measure the relative contribution of larger lexical families or individual morphological classes. Removing all one- and two-character roots is a deliberately blunt diagnostic test. It does not tell us how much of the remaining performance depends on particular families such as *cortex*, *suber*, *exterior*, or *annulus*, nor does it isolate the contribution of prefixes, suffixes, environmental terms, structural terms, or functional markers. Future work should therefore include component-level ablation, family-level sensitivity analysis, and larger lexical-unit evaluation. Such experiments would test whether the model remains stable when individual semantic families are removed, when specific morphological classes are withheld, or when longer roots are evaluated independently of shorter forms.

6.5 Model-Driven Evaluation and Reproducibility

The principal methodological contribution of this work is the use of a model-driven evaluation framework rather than a sequence of isolated translations. The parser is not merely a tool for producing readings; it is the software implementation of the model used to test the hypothesis. Its assumptions are encoded in *ROOTS*, *PREFIXES*, *SUFFIXES*, token cleaning, section classification, and strict parsing logic. Because these assumptions are explicit, they can be inspected, rerun, modified, and challenged. This is essential for a field in which many proposed solutions have depended on selective examples, private procedures, or interpretive flexibility.

Model-driven evaluation also changes the status of failure. In a conventional interpretive essay, unresolved tokens may disappear from view. In the v22 framework, failure cases are exported. Every evaluated token is assigned a strict parser status, and unresolved tokens are listed with a failure reason. This does not make the parser stronger in a rhetorical sense; it makes it more accountable. A model that exposes its failures can be improved systematically. A model that hides its failures can only be accepted or rejected impressionistically.

The same principle applies to reproducibility. The parser exports corpus performance, section statistics, permutation statistics, representative parses, lexical stability, and failure cases as machine-readable tables. It also exports token-level supplementary data for all 31,007 evaluated tokens. This allows readers to verify whether a reported number is derived from the code rather than inserted into the paper manually. It also allows future parser versions to be compared against v22. If a later version improves strict coverage, it must be possible to ask whether the improvement came from a principled lexical refinement, a change in tokenization, a new compound rule, an expanded root inventory, or a relaxation of evaluation criteria.

Version-controlled methodology is particularly important here. The v22 parser is not presented as a timeless solution, but as a reproducible state of a developing model. Earlier exploratory versions produced broader recognition figures, including the legacy 98.66% recognition rate. The v22 update separates that exploratory recognition from strict full-token validation. Later versions may add compound rules, refine lexical families, or test larger morphological units, but they should do so as explicit versioned changes. This is how cumulative scientific refinement becomes possible. A claim can be corrected without erasing the previous result; a new experiment can be added without retroactively changing the baseline; a failure can become a target for the next model rather than a reason to obscure the current one.

This is also why the paper distinguishes official parser outputs from diagnostic experiments. The official v22 strict result remains 15,848 tokens, or 51.11%. The ablation results do not replace that number. They explain its dependence on components of the lexical inventory. Similarly, the Exploratory Recognition Mode remains useful for tracing possible internal structure, but it is not the primary validation criterion. The point of the version-controlled approach is to preserve these distinctions rather than collapse them into a single headline figure.

6.6 Cumulative Scientific Refinement

The development from exploratory recognition to strict validation illustrates a broader principle: progress in the analysis of the Voynich Manuscript should be cumulative, auditable, and reversible. A hypothesis should not become stronger by accumulating untracked exceptions. It should become stronger by surviving increasingly explicit tests. In the present work, the stricter v22 parser reduced the headline coverage from the legacy exploratory figure to a more conservative 51.11%, but the reduction improved interpretability. It made clear what the parser can do under a full-token criterion and what remains unresolved.

The same refinement is visible in the treatment of lexical families. Rather than using semantic flexibility to explain difficult contexts, v22 fixes lexical values and reports whether variants are reused consistently across sections. This does not guarantee correctness, but it gives the model a stable surface against which criticism can operate. A proposed family may be too broad, too narrow, or historically implausible. The important point is that it can be identified, changed, and retested. This makes the work cumulative rather than merely persuasive.

The ablation experiments extend the same logic. They do not improve the parser's score. They make the score more interpretable. By showing that removal of one-character roots reduces strict coverage to 38.82%, and removal of one- and two-character roots reduces strict coverage to 20.37%, the analysis clarifies both the strength and vulnerability of the model. The strength is that the baseline is not

indistinguishable from random lexical inventories and that some strict coverage remains even under severe ablation. The vulnerability is that short roots play a major role and must be examined carefully in future versions.

This kind of refinement is especially important because the Voynich Manuscript has a long history of overclaiming. A responsible computational discussion must therefore resist the temptation to convert partial structure into final decipherment. The present parser may be detecting real morphological regularities, regularities induced by transcription conventions, regularities produced by scribal generation, or some combination of these. The statistical tests narrow the space of plausible explanations, but they do not close it.

6.7 Historical Interpretation Remains Open

The results are compatible with the hypothesis that the manuscript encodes a compressed botanical-medical system, but they do not compel that conclusion. The lexical inventory was designed around botanical, anatomical, and pharmaceutical values, and several section-level patterns are consistent with that design. Nevertheless, historical interpretation requires more than internal parser performance. It requires independent support from palaeography, codicology, iconography, comparative manuscript traditions, and historical linguistics. Those forms of evidence are relevant to the broader argument, but they should not be treated as already settled by the parser.

The relationship between morphology and illustration therefore remains a field for further testing. The parser operates on EVA transcription and section labels, not on images. Any correspondence between parsed values and visual content should be treated as an external consistency check rather than a built-in premise of the algorithm. This is particularly important for the pharmaceutical material, where the traditional visual classification and the parser's internal subsections do not serve the same analytical purpose. Morphological results can inform iconographic interpretation, but they should not be allowed to replace it.

The present discussion therefore stops short of final synthesis. It argues that parser version v22 supports a reproducible framework for evaluating the proposed morphological hypothesis and that the current evidence supports non-random structure under conservative validation. It also identifies specific dependencies, especially the contribution of short lexical roots, that require further testing. Whether these structures ultimately support the proposed historical reading of the manuscript is a question reserved for Chapter 7 and for future independent replication.

7. Conclusion

This study investigated whether the Voynich Manuscript can be examined through a reproducible morphological model rather than as an undecipherable sequence of isolated symbols. Instead of seeking a definitive historical decipherment through selected translations, it developed an explicit computational framework in which a fixed lexical inventory, predefined morphological rules, and conservative validation criteria were applied consistently across the complete corpus. The objective was to determine whether the proposed hypothesis could explain corpus-wide textual structure under reproducible conditions.

To address this question, parser version v22 combined a documented lexical knowledge base, fixed lexical-family definitions, Strict Full-Token Validation, permutation testing, section-level evaluation, and robustness analyses based on progressive short-root ablation. Together, these components established a reproducible evaluation pipeline capable of testing a corpus-wide morphological hypothesis under explicit computational constraints.

The results indicate that parser version v22 explains a substantial proportion of the corpus while remaining statistically distinct from randomized lexical organization. Strict Full-Token Validation successfully parsed 15,848 of 31,007 tokens (51.11%). Randomized lexical inventories produced a null mean of 2.112%, yielding a separation of 30.07 standard deviations (empirical $p < 0.0001$). Robustness analyses further demonstrated that strict coverage remained 38.82% after removing all one-character lexical roots and 20.37% after removing every one- and two-character lexical root, indicating that short lexical elements contribute substantially but do not solely account for the observed corpus-wide structure.

These findings should be interpreted cautiously. They do not demonstrate that the Voynich Manuscript has been fully deciphered, nor do they establish that the proposed Latin-derived lexical assignments necessarily reflect the manuscript's original historical language. Rather, they show that the proposed model captures reproducible, statistically non-random morphological organization that can be evaluated objectively under fixed computational rules.

The broader contribution of this study is methodological as well as linguistic. By integrating an explicit parser implementation, documented lexical families, conservative validation, permutation testing, and robustness analysis within a version-controlled framework, the study establishes a reproducible baseline against which competing morphological hypotheses may be compared empirically. The value of the framework therefore lies not only in the particular lexical inventory proposed here, but also in the transparent evaluation methodology that accompanies it.

Because the parser implementation, exported datasets, and evaluation procedures are publicly available, future investigations can directly reproduce, refine, extend, or challenge the present framework under identical computational conditions. This enables subsequent work to build upon a common experimental foundation rather than beginning from independent and incomparable decipherment proposals.

The significance of the present study therefore extends beyond the specific Latin-derived morphological framework advanced here. Whether future investigations ultimately confirm, revise, or reject the proposed interpretation, they can now evaluate alternative hypotheses against an explicit computational and statistical baseline. By shifting Voynich research from isolated interpretative claims toward cumulative empirical comparison, the framework introduced in this study provides a foundation upon which future progress can be measured objectively, reproduced independently, and refined through transparent scientific evaluation.

Declarations and References

Acknowledgements

The author gratefully acknowledges the publicly available manuscript images and scholarly resources that have supported Voynich Manuscript research over many decades.

Funding

The author received no external funding for this research.

Competing Interests

The author declares that there are no competing interests.

Author Contributions

Uk Kim conceived the study, developed the computational framework and parser, performed all analyses, interpreted the results, and wrote, reviewed, and approved the final manuscript.

AI Assistance Statement

During preparation of this manuscript, the author used OpenAI ChatGPT, OpenAI Codex, and Anthropic Claude Sonnet 4.6 as language-assistance and software-development tools. These systems assisted with manuscript drafting, language editing, structural editing, parser implementation, verification

experiments, and preparation of supplementary materials under the author's direction. All research questions, lexical definitions, parser architecture, computational methodology, statistical interpretation, experimental decisions, and final editorial judgments were determined and verified by the author. No AI system generated the scientific conclusions or determined the interpretation presented in this study. The author accepts full responsibility for the content of this manuscript.

Data Availability

The parser outputs, lexical resources, evaluation datasets, supplementary tables, and robustness analyses used in this study are available through the project GitHub repository. A permanent archived version corresponding to this manuscript is available through Zenodo (DOI: 10.5281/zenodo.20822744). Future software development may continue through the GitHub repository.

Code Availability

The complete parser implementation used in this study, including parser version v22, analysis scripts, lexical resources, and evaluation utilities, is available through the project GitHub repository. The manuscript should be evaluated against the archived Zenodo snapshot corresponding to parser version v22, while the GitHub repository may contain later development versions.

References

- Beinecke Rare Book & Manuscript Library. (n.d.). *The Voynich Manuscript (Beinecke MS 408)*. Yale University Library. <https://collections.library.yale.edu/catalog/2002046>
- Bowern, C., & Lindemann, L. (2021). The linguistics of the Voynich Manuscript. *Annual Review of Linguistics*, 7, 285–308. <https://doi.org/10.1146/annurev-linguistics-011619-030613>
- Landini, G., & Zandbergen, R. (n.d.). Text analysis: Transliteration of the text (EVA). Voynich.nu. Retrieved July 1, 2026, from <https://www.voynich.nu/transcr.html>
- Montemurro, M. A., & Zanette, D. H. (2013). Keywords and co-occurrence patterns in the Voynich manuscript: An information-theoretic analysis. *PLOS ONE*, 8(6), e66344. <https://doi.org/10.1371/journal.pone.0066344>
- Rugg, G. (2004). The mystery of the Voynich manuscript. *Scientific American*, 291(1), 104–109.
- Schinner, A. (2007). The Voynich manuscript: Evidence of the hoax hypothesis. *Cryptologia*, 31(2), 95–107. <https://doi.org/10.1080/01611190601133539>

University of Arizona. (2011, February 9). *UA experts determine age of book nobody can read*.
University of Arizona News.

Zandbergen, R. (2025). *The Voynich Manuscript*. Voynich.nu. <https://www.voynich.nu/>